Definition and Pathogenesis of Atopy

Atopy can be defined as a familial hypersensitivity of the skin and mucosa to environmental substances, associated with increased production of immunoglobulin E (IgE) or altered pharmacologic reactivity. The ETFAD defines atopy as the "familial tendency to develop Th2 responses against common environmental antigens," encompassing both IgE-associated (extrinsic) and non-IgE-associated (intrinsic) subtypes within the atopic spectrum.

Genetic and Molecular Basis

Atopic diseases are genetically linked, with a concordance rate of 80% in monozygotic twins compared to 30% in dizygotic twins. Genetic polymorphisms involved in atopic dermatitis (AD) affect mediators of atopic inflammation located on various chromosomes, some of which are also implicated in respiratory atopy. The strongest genetic association is with mutations in the *filaggrin* gene, which is also linked to ichthyosis vulgaris, underscoring the role of epidermal barrier dysfunction in AD. Clinically, concomitant ichthyosis vulgaris may be identified by increased palmar hyperlinearity, a useful indicator of filaggrin gene mutations.

Clinical Presentation

In early infancy, AD may present as "cradle cap"—a yellowish, seborrheic, crusted scaling on the scalp—which typically responds to emollient therapy alone. The disease usually begins on the face (with relative sparing of the middle face), extensor surfaces of the limbs, and trunk, often with oozing and crusting. As the child grows, involvement shifts to the flexural areas, neck, and hands, accompanied by dry skin.

Skin barrier dysfunction in AD is evidenced by increased transepidermal water loss (TEWL). Measuring TEWL with an evaporimeter and discussing results with patients can enhance adherence to emollient therapy. Lichenification results from chronic scratching and rubbing, often occurring at night. The prurigo variant of AD is characterized by excoriated, nodular, and lichenified lesions.

Exacerbations typically begin with intense pruritus in the absence of visible lesions, followed by erythema, papules, and infiltration. Histopathologic findings are not specific for AD; thus, routine biopsy is not diagnostic but may help exclude conditions such as cutaneous T-cell lymphoma in adults.

Diagnosis

In the absence of a specific laboratory marker, AD is diagnosed clinically. While an experienced clinician's judgment surpasses available diagnostic criteria, standardized criteria are essential for epidemiological studies and clinical trial enrollment. Most criteria focus on cutaneous signs of atopy.

- **Hanifin and Rajka criteria** require three of the following four: pruritus, typical morphology and distribution, chronic or relapsing course, and personal or family history of atopy. Additionally, at least three minor criteria from a list of 21 must be fulfilled.
- **UK Working Party criteria** include an itchy skin condition in the past 12 months plus at least three of the following: onset before age two, history of skin fold involvement, generalized dry skin, personal or family history of other atopic diseases, and visible flexural eczema.

Impact on Quality of Life

Atopic dermatitis significantly affects all aspects of a patient's life. Moderate-to-severe AD is associated with depression, anxiety, reduced quality of life, impaired self-esteem, and increased suicide risk. The social burden of moderate-to-severe AD is comparable to that of psoriasis, chronic obstructive pulmonary disease, cardiovascular disease, and diabetes.

Management of Flares

Managing acute exacerbations (flares) of AD is challenging, requiring rapid symptom control without disrupting long-term management goals such as flare prevention and minimizing treatment side effects. Flares may reveal underlying triggers, including allergies or infections. Therefore, initial evaluation should include a detailed history of flare circumstances and a thorough dermatologic examination, including assessment of lymph nodes, orifices, and skin folds. A professional and empathetic approach during flare management lays the foundation for future treatment adherence.